

Implementation map of the DY0 and DY⁺ finite remainders

(non-singlet “S” sector) and the Mathematica cross-checks

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This note documents, term by term, *where* each finite-remainder piece of the neutral-current (DY0) and charged-current (DY⁺, W⁺) mixed QCD⊗EW NNLO subtraction is implemented in the Fortran code, and which of the DY⁺ pieces were checked symbolically against the Mathematica notebook. The analytic definitions are the cells FINITEDY0[1..7] of math/DY0_qq.w1 and FINITEDYp[1..7], FINITEDYp[60] of math/DY+_qq.w1.

1 Conventions and switch

The neutral/charged current is selected at compile time by the preprocessor flag `_Vcharge`: 0 = DY0 ($q\bar{q} \rightarrow \gamma^*/Z$), +1 = W⁺, -1 = W⁻. It drives `current_charge` ∈ {0, ±1} (`src/Parms/mod_proc_parms.f90`). All CC routines follow the convention: implement the `_Vcharge==0` branch (DY0) first, then `_Vcharge==+1` (W⁺); `_Vcharge==−1` (W⁻) currently executes a `stop` (“not yet implemented”). Only W⁺ is implemented.

Throughout, $s = 4E_C^2 = \hat{s}$, $L_{\mu s} = \ln(\mu^2/s) = \ln(\mu^2/4E_C^2)$, $D_k = [\ln^k(1-z)/(1-z)]_+$, and $E_C = M_V/2$. The two recurring z -brackets are

$$B_1(z) = 1 - z + 4D_1 + (1 + z - 2D_0)L_{\mu s} - \frac{2\ln z}{1-z} + (1 + z)(-2\ln(1-z) + \ln z), \quad (1)$$

$$B_A(z) = 1 - z - 2(1 + z)\ln(1-z) + 4D_1 + (1 + z - 2D_0)\ln\frac{\mu^2}{4E_C^2}, \quad (2)$$

$$B_B(z) = 1 - z - 2(1 + z)\ln(1-z) + (1 + z - 2D_0)\ln z + 4D_1 + (1 + z - 2D_0)L_{\mu s}. \quad (3)$$

B_1 is the bracket of FINITEDY0[1], FINITEDYp[1], FINITEDY0[3], FINITEDYp[3] (they all share it). B_A is the collinear splitting of the single-boost “term A” ([6]/[60]); B_B multiplies its “term B”.

2 Building blocks (both channels)

Object	File	Meaning / notes
res_tree_qqb res_tree_qqb_gen	mod_amplitudes_- tree_pp11.f90	Born FLM. NC form returns (2, 2) [dn,up]; <code>_gen</code> returns the (−5:7, −5:7) flavour matrix and is CC-aware (selects entries with $Q_{\text{IS}}(i) + Q_{\text{IS}}(j) = \text{current_charge}$).
res_qcdloop_qqb res_qcdloop_qqb_gen	mod_amplitudes_- loop_pp11.f90	QCD one-loop finite $\text{FLV}_{\text{QCD}} = -8C_F$ Born. NC (2, 2) / CC (−5:7, −5:7).
res_ewkloop_qqb res_ewkloop_qqb_gen	mod_amplitudes_- loop_pp11.f90	EW one-loop finite FLV_{EW} (OpenLoops). NC (2, 3) [dn,up,b] / CC (−5:7, −5:7); carries the electric charges internally.
multiply_IS_charges_- sq(r,leg)	mod_aux_- sectors.f90	$r(i, j) \rightarrow r(i, j) Q_{\text{IS}}(i)^2$ (leg=1) or $Q_{\text{IS}}(j)^2$ (leg=2).

Object	File	Meaning / notes
get_respdf_hoppet get_respdf_hoppet_gen	mod_aux_- sectors.f90	PDF convolution with pre-boosted pdf_- tables (xPij/PDFs); NC uses an explicit luminosity, _gen uses gen_lumi on the full flavour matrix.
get_respdf_gen_mu	mod_aux_- sectors.f90	convolves an amplitude already multiplied by a μ -dependent integrated-subtraction matrix $(-5:7, -5:7, i_{\text{pdf}})$.
fin_ns_vqcd (l. 26)	mod_int_sub_- nnlo.f90	DY0 elastic QCD I -operator: $[4(\text{Li}_2(\bar{\eta}_{13}) - \text{Li}_2(\bar{\eta}_{14})) - 6 \ln(\bar{\eta}_{13}/\bar{\eta}_{14})] Q_q Q_\ell + (13 - 4\zeta_2) Q_\ell^2$.
calG_QqQ1 (l. 87), calG_QqQ1_L (l. 140)	mod_int_sub_- nnlo.f90	DY0 soft function $G_q[Q_q, Q_e]$ and the coef- ficient of $L_{\mu s}$ in [7] (the latter <i>excludes</i> the Q_q^2 piece).
calG_ONLOQCD_ns (l. 276)	mod_int_sub_- nnlo.f90	the <i>real-emission</i> NLO-QCD soft (used by the $V+g$ sectors). Not equal to genGnloQCD – see §6.
genGnloQCD_cc (l. 416), legfinalcalG_cc (l. 458)	mod_int_sub_- nnlo.f90	CC elastic soft = notebook genGnloQCD (validated), and legfinalcalG_- cc(..., i_{leg}) = genGnloQCD $- 3Q_i^2 \ln(4E_C^2/\mu^2)$ = notebook leg1/leg2finalcalG.
fin_ns_vqcd_cc (l. 337)	mod_int_sub_- nnlo.f90	DY ⁺ elastic QCD I -operator, FINITEDYp[2] (with the Q^2 shift, §5).
fin_ns_v7_cc (l. 484)	mod_int_sub_- nnlo.f90	DY ⁺ elastic soft, FINITEDYp[7].
mySub_ns / mySub_ns_Lmu	mod_hoppet_- nnlo.f90	kernels filling xPij/xPij_Lmu for the s_- v sector (vqcd, vewk); B_1 -type. $\delta(1-z)$: $2\zeta_2 / -3/2$ (“half per leg”).
mySub_TCq / _Lmu / _Lmu2	mod_hoppet_- nnlo.f90	kernels filling xPij/xPij_Lmu/xPij_Lmu2 for the s_12 sector; the big Q_q^2 z -space bracket of [6]. $\delta(1-z)$: $-\frac{2\pi^4}{45}, -2(\pi^2+8\zeta_3),$ $\frac{9}{2} - 4\zeta_2$.
xPij_{A,B,C,D}_{1,2}	mod_hoppet_- nnlo.f90	the four double-boost tables AA/BB/CC/DD of [5] (kernel mySub_qq).

All three finite-remainder integrands live in `src/Sectors/mod_xsects_nnlo_s_ns.f90`: `xsect_nnlo_s_ns_vqcd` (l. 480) $\rightarrow$ [1], [2]; `xsect_nnlo_s_ns_vewk` (l. 605) $\rightarrow$ [3], [4]; `xsect_nnlo_s_ns_s12` (l. 83) $\rightarrow$ [5], [6], [60], [7]. The `s12` routine fills three histogram slots: `kin(1)` (elastic point, tables `respdf_1..4`), `kin(2)` $= F[z_1 p_1, p_2]$ (boosted leg 1), `kin(3)` $= F[p_1, z_2 p_2]$ (boosted leg 2).

3 DY0 (neutral current): where each term is

Term	Location	Implementation
[1]	vqcd: respdf_2, respdf_3	$Q_q^2(\text{FLV}_{\text{QCD}}[z p_1, p_2] + \text{FLV}_{\text{QCD}}[p_1, z p_2]) B_1$. <code>res_qcdloop_qqb</code> $\rightarrow$ <code>FLV_QCD</code> ; charges $Q_{\text{dn}}^2/Q_{\text{up}}^2$; then $F[z_1, \cdot] = \text{get_respdf_hoppet}(\text{xPij}, \text{PDFs}, \dots, [\text{xPij_Lmu}])$ and $F[\cdot, z_2] = (\text{PDFs}, \text{xPij})$. Kernel <code>mySub_ns</code> .
[2]	vqcd: respdf_1 and δ of respdf_2/3	$(13 - \frac{2\pi^2}{3})Q_e^2 + \frac{2\pi^2}{3}Q_q^2 - 3Q_q^2 L_{\mu s} + Q_e Q_q(\dots)$. The Q_e^2 and $Q_e Q_q$ parts = <code>fin_ns_vqcd</code> $\otimes$ <code>FLV_QCD</code> (<code>respdf_1</code> , elastic (PDFs, PDFs)). The $\frac{2\pi^2}{3}Q_q^2$ and $-3Q_q^2 L_{\mu s}$ parts come from the $\delta(1-z)$ of <code>mySub_ns/mySub_ns_Lmu</code> ($2\zeta_2, -3/2$; per leg, $\times Q_q^2$) delivered inside <code>respdf_2/3</code> .

Term	Location	Implementation
[3]	vewk: respdf_1, respdf_2	$C_F(\text{FLV}_{\text{EW}}[zp_1, p_2] + \text{FLV}_{\text{EW}}[p_1, zp_2]) B_1$. res_ewkloop_qqb $\rightarrow$ FLV_{EW} ; <i>no</i> charge weighting (FLV_{EW} carries the charges); ns_lumi_splitb ; overall $\times C_F$. Kernel mySub_ns .
[4]	vewk: δ of respdf_1/2	$C_F(\frac{2\pi^2}{3} - 3L_{\mu s})\text{FLV}_{\text{EW}}$, from the mySub_ns/_Lmu $\delta(1-z)$ summed over the two legs (no charge weighting), $\times C_F$.
[5]	s12: kin(1), respdf_1	$2C_F Q_q^2 \text{FLM}[z_1 p_1, z_2 p_2](\text{AA}+\text{BB}+\text{CC}+\text{DD})$. respdf_1 = $2 \sum_X \text{get_respdf_hoppet}(\text{xPij_X_1}, \text{xPij_X_2}, \dots)$, $X \in \{A, B, C, D\}$; res_tree_qqb $\times Q_{\text{dn}}^2/Q_{\text{up}}^2$; $\times C_F$.
[6]	s12: kin(2) + respdf_2,3,4 of kin(1)	Leg-1 single boost, three pieces: $A (G_q \cdot B_A)$: kin(2) = $C_F S_1 \text{calG_QqQ1_FLM}$ with $S_1 = \text{PqqNLO_1}(0) + \text{Pqq0_R_1}(0)L_{\mu s}$; its plus-endpoint is subtracted in kin(1) respdf_3 = $(S_1^+ + S_2^+) \text{calG_QqQ1FLM}$, $S_i^+ = -(\text{PqqNLO}(1) + \text{Pqq0_R}(1)L_{\mu s})$. $B (-2Q_e Q_q B_B \ln z \ln \frac{s_{24}}{s_{23}})$: kin(1) respdf_2 , get_respdf_hoppet (xPij_B_1 , PDFs, ...) $\times Q_e Q_q \times 2 \ln(u/t)$. $C (Q_q^2 \times \text{big } z\text{-space bracket})$: kin(1) respdf_2 , get_respdf_hoppet (xPij , PDFs, ..., [xPij_Lmu , xPij_Lmu2]) $\times Q_q^2$ (kernel mySub_TCqq). (Leg 2: kin(3) for A, and the leg-2 halves of respdf_2 .)
[7]	s12: kin(1), respdf_4 + δ of term C	$C_F(4\zeta_2 - 3 \ln \frac{\mu^2}{4E_C^2}) G_q \text{FLM} + C_F Q_q^2 (-\frac{4\pi^4}{45} - (2\pi^2 + 32\zeta_3)L_{\mu s} + (9 - \frac{4\pi^2}{3})L_{\mu s}^2) \text{FLM}$. respdf_4 = $4\zeta_2 \text{calG_QqQ1FLM} + \text{calG_QqQ1_L} \cdot L_{\mu s} \text{FLM}$; the pure- $Q_q^2 (\pi^4, L_{\mu s}, L_{\mu s}^2)$ part is supplied by the $\delta(1-z)$ of mySub_TCqq/_Lmu/_Lmu2 inside term C (§5).

4 DY⁺ (charged current, W⁺): where each term is

All entries are the **_Vcharge==+1** branches of the same three routines. Leptonic ids are set to [0,0,**id_nue**,**-id_el**] (νe^+); the charged lepton is leg 4 (**ilept=4**, **Qlept=+1**). Amplitudes use the **_gen** (-5:7, -5:7) forms; per-leg electric charges are applied by **multiply_IS_charges_sq**.

Term	Location	Implementation (differences vs DY0)
[1]	vqcd: respdf_2, respdf_3	$(Q_{q'}^2 \text{FLV}_{\text{QCD}}[q, q'z] + Q_q^2 \text{FLV}_{\text{QCD}}[qz, q']) B_1$. res_vqcdloop_qqb_gen $\rightarrow$ FLV_{QCD} ; <i>per-leg</i> charge: $F[z_1, \cdot] = \text{get_respdf_hoppet_gen}(\text{xPij}, \text{PDFs}, \dots)$ on multiply_IS_charges_sq ($\cdot, 1$) (Q_q^2), $F[\cdot, z_2]$ on multiply_IS_charges_sq ($\cdot, 2$) ($Q_{q'}^2$). Same xPij (mySub_ns).
[2]	vqcd: respdf_1	fin_ns_vqcd_cc $\otimes$ get_respdf_gen_mu (elastic). The full charge-quadratic G_p (in $Q_{\ell'}^2, Q_{\ell'} Q_q, Q_{\ell'} Q_{q'}, Q_q Q_{q'}$) <i>minus</i> the spurious per-leg Q^2 shipped by the reused xPij δ (§5).
[3]	vewk: respdf_1, respdf_2	$C_F(\text{FLV}_{\text{EW}}[qz, q'] + \text{FLV}_{\text{EW}}[q, q'z]) B_1$. res_ewkloop_qqb_gen $\rightarrow$ FLV_{EW} ; get_respdf_hoppet_gen , $\times C_F$. <i>No</i> charge weighting and <i>no</i> Q^2 shift (identical structure to DY0 [3]).

Term	Location	Implementation (differences vs DY0)
[4]	vewk: δ of respdf_1/2	$C_F(\frac{2\pi^2}{3} - 3L_{\mu s})\text{FLV}_{\text{EW}}$ from the xPij δ ; identical to DY0 [4], no compensation needed (no per-leg Q^2 weighting).
[5]	s12: kin(1), respdf_1	$C_F(Q_q^2 + Q_{q'}^2)\text{FLM}[z_1 p_1, z_2 p_2](\text{AA} + \dots + \text{DD})$. res_tree_qqb_gen $\rightarrow$ FLM; the $(Q_q^2 + Q_{q'}^2)$ prefactor = multiply_IS_charges_sq ($\cdot, 1$)+($\cdot, 2$) (the two QCD $\leftrightarrow$ QED leg assignments); $\sum_X \text{get_respdf_hoppet_gen}(\text{xPij_X_1}, \text{xPij_X_2}, \dots)$, <i>no</i> factor 2.
[6]	s12: kin(2) + respdf_2,3,4	Leg-1 single boost: A: kin(2) = $C_F S_1 \text{legfinalcalG_cc}(\cdot, 1)\text{FLM}$ (via get_respdf_gen_mu); plus-endpoint in kin(1) respdf_3 = $S_1^+ \text{legfinalcalG_cc}(\cdot, 1) + S_2^+ \text{legfinalcalG_cc}(\cdot, 2)$. B $(-2Q_{\ell'} Q_q \ln z \ln \frac{s_{14}}{s_{12}})$: kin(1) respdf_2, get_respdf_hoppet_gen (xPij_B_1, PDFs, ...) on FLM $\cdot Q_{\text{IS}}(al)$ (first power), $\times (-2Q_{\ell'} \ln \frac{s_{14}}{s_{12}})$. C $(Q_q^2 \times \text{big bracket})$: kin(1) respdf_2, get_respdf_hoppet_gen (xPij, PDFs, ..., [xPij_Lmu, xPij_Lmu2]) on multiply_IS_charges_sq ($\cdot, 1$) (Q_q^2).
[60]	s12: kin(3) + leg-2 halves	Leg-2 mirror of [6]: kin(3) = $C_F S_2 \text{legfinalcalG_cc}(\cdot, 2)\text{FLM}$; term B $\rightarrow$ (PDFs, xPij_C_2) on FLM $\cdot Q_{\text{IS}}(be)$, $\times (-2Q_{\ell'} \ln \frac{s_{24}}{s_{12}})$; term C $\rightarrow$ multiply_IS_charges_sq ($\cdot, 2$) ($Q_{q'}^2$).
[7]	s12: kin(1), respdf_4 + δ of term C	fin_ns_v7_cc $\otimes$ get_respdf_gen_mu : $(4\zeta_2 - 3L_{\mu s})\text{genGnloQCD_cc} + (\text{EB} - \delta_C)$, where EB is the explicit charge bracket of FINITEDYp[7] and δ_C removes the pure- Q^2 elastic already shipped by term C's xPij δ (§5).

5 The $\delta(1-z)$ book-keeping (“ Q^2 shift”)

The single/double-boost kernels reused from DY0 carry a $\delta(1-z)$ endpoint that, once weighted per initial leg by **multiply_IS_charges_sq** ($\times Q_{\text{IS}}^2$), delivers elastic contributions. For DY0 these are exactly the pure- Q_q^2 pieces of [2] (from **mySub_ns**) and [7] (from **mySub_TCqq**); this was verified directly against the notebook. For DY⁺ the situation differs term by term:

- **vqcd** / FINITEDYp[2]: G_p has *no* pure $Q_q^2/Q_{q'}^2$ term, so the $2\zeta_2 / -3/2$ deltas of **mySub_ns/_Lmu** (times Q_{IS}^2 per leg) are *spurious*. **fin_ns_vqcd_cc** subtracts them via **coeff_Qsq** = $-2\zeta_2 + \frac{3}{2}L_{\mu s}$ on $(Q_{\text{IS}}(al))^2 + Q_{\text{IS}}(be)^2$.
- **vewk** / FINITEDYp[4]: no per-leg charge weighting, so the deltas reproduce $C_F(\frac{2\pi^2}{3} - 3L_{\mu s})\text{FLV}_{\text{EW}}$ exactly (as in DY0); *no* compensation.
- **s12** / FINITEDYp[7]: [7] *does* contain pure $Q_q^2/Q_{q'}^2$, and the **mySub_TCqq** deltas ($-\frac{2\pi^4}{45}$, $-2(\pi^2 + 8\zeta_3)$, $\frac{9}{2} - 4\zeta_2$; per leg, $\times Q_{\text{IS}}^2$) supply them. **fin_ns_v7_cc** therefore *subtracts* exactly that combination (δ_C) from the explicit bracket EB so that **respdf_4** + (term-C δ) = the notebook [7].

The leg- i soft function used in [6]/[60] term A and in respdf_3 is **legfinalcalG_cc**(..., i) = **genGnloQCD** + $3 Q_{\text{IS}}(\text{leg } i)^2 L_{\mu s}$ = **genGnloQCD** - $3Q_i^2 \ln(4E_C^2/\mu^2)$.

6 Cross-checks against the Mathematica notebook

The script `math/validate_DYp7.wl` performs the symbolic checks below. It is run in a kernel that has first evaluated all input cells of `DY+_qq.wl` (so that `genGnloQCD`, `FINITEDYp[7]`, `leg1finalcalG`, `leg2finalcalG` are defined). The runner (`run_validate.wls`, invoked with `wolframscript -file`) does

```
Get["DY+_qq.wl"] → Cases[... , Cell[BoxData[b], "Input"] :> x] → ToExpression → Get["validate_DYp7.wl"]
```

The Fortran expressions are transcribed with the map $Q_q \rightarrow Q_{IS}(al)$, $Q_{q'} \rightarrow Q_{IS}(be)$, $Q_{\ell'} \rightarrow Q_{\text{lept}}$, $L \equiv \ln(\mu^2/4E_C^2)$, $\ell_E \equiv \ln(E_C/E_4)$, and the notebook is mapped to the same variables via its own leg-replacement rules $\ln(\text{twoEConMu}) \rightarrow -L/2$, $\ln(\text{twoE4onMu}) \rightarrow -\ell_E - L/2$, $\eta_{12} \rightarrow 1$.

Check	What it compares	Result
1	$[(4\zeta_2 - 3L) \text{genGnloQCD} + \text{EBcode}] - \text{FINITEDYp}[7]$ (the <code>genGnloQCD</code> cancels, so this isolates the explicit-bracket transcription EB and the δ_C subtraction).	= 0 PASS
2	the Fortran <code>genGnloQCD_cc</code> charge-quadratic coefficients – the notebook <code>genGnloQCD</code> (mapped).	= 0 PASS
3	<code>legfinalcalG_cc</code> (·, 1) – notebook <code>leg1finalcalG</code> .	= 0 PASS
4	<code>legfinalcalG_cc</code> (·, 2) – notebook <code>leg2finalcalG</code> .	= 0 PASS

Important finding. Check 2 exposed that the real-emission soft `calG_ONLOQCD_ns` is *not* equal to the elastic soft `genGnloQCD` (it carries an extra $-6\ln(E_C/E_4)$ in the $Q_q Q_{q'}$ coefficient and sign-flipped η -terms). An earlier draft of the DY^+ [6]/[60] term A and [7] that reused `calG_ONLOQCD_ns` was therefore incorrect; both now use `genGnloQCD_cc/legfinalcalG_cc`, which pass Checks 2–4.

Coverage. The checks validate, symbolically, the pieces that are pure integrated-subtraction functions: `FINITEDYp[7]` (Check 1), the CC soft `genGnloQCD` (Check 2) and the leg- i soft functions of [6]/[60] term A (Checks 3–4). The remaining DY^+ pieces ([1]–[5], terms B/C of [6]/[60]) reuse the *same* DY0 HOPPET tables (`xPij`, `xPij_X`) with per-leg charges; their correctness rests on the DY0 implementation (checked against `oldcode`) plus the charge/kinematic factors given in §4, and is not re-derived symbolically here.

7 Status

- **Implemented and (where symbolic) validated for W^+ :** `FINITEDYp[1..7]` and `FINITEDYp[60]`.
- **Not implemented:** W^- (`_Vcharge==1`) – every routine executes a `stop`.
- The secondary double-real flavour channels (`_qq`, `_qqb`) are separate sectors and are *not* part of `FINITEDYp[1..7,60]`.